

**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 2- MCQ TEST**

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| <b>Course Code:</b>   | <b>CSA12</b>                                                                                                                                                                                                                                                                       | <b>Course Title:</b> | <b>Computer Architecture</b>           |
| <b>CO Assessed:</b>   | <b>CO3</b>                                                                                                                                                                                                                                                                         | <b>Assessment:</b>   | <b>ASSESSMENT TOOL 2-<br/>MCQ TEST</b> |
| <b>Weightage:</b>     | <b>25%</b>                                                                                                                                                                                                                                                                         | <b>Total Marks:</b>  | <b>25</b>                              |
| <b>CO3 Statement:</b> | <i>Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT). (BL6)</i> |                      |                                        |
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| <b>Department:</b>    | <b>CSE-AI</b>                                                                                                                                                                                                                                                                      | <b>Date:</b>         | <b>30/07/2026</b>                      |

**ANNEXURE A**

1. A processor transfers data between registers before performing an arithmetic operation in the ALU. Which component primarily controls these register transfer operations?
  - a) Arithmetic Logic Unit
  - b) Control Unit
  - c) Memory Unit
  - d) Cache Controller
2. In a system where multiple registers are used for intermediate storage, data must be efficiently moved before computation. Which concept ensures correct sequencing of such operations?
  - a) Register transfer language
  - b) Instruction pipelining
  - c) Cache mapping
  - d) Interrupt handling
3. A processor performs logical AND, OR, and XOR operations on data stored in registers. These operations are handled by which unit?
  - a) Control Unit
  - b) ALU
  - c) Bus Interface Unit
  - d) Memory Management Unit
4. A system designer chooses a single-bus architecture for simplicity, but performance issues arise when multiple components attempt data transfer simultaneously. What is the primary limitation here?
  - a) Lack of memory
  - b) Bus contention
  - c) High clock speed
  - d) Complex control logic
5. In a multi-bus organization, data transfer between CPU and memory is faster compared to a single-bus system. What is the main reason for this improvement?
  - a) Reduced instruction size
  - b) Parallel data transfer capability
  - c) Increased clock frequency
  - d) Simplified hardware

6. A processor executes instructions using a 5-stage pipeline. If one stage takes significantly longer than others, what will be the impact?
  - a) Increased throughput
  - b) Pipeline stalls
  - c) Reduced instruction size
  - d) Faster execution
7. During pipelined execution, an instruction depends on the result of a previous instruction that has not yet completed. What type of hazard does this represent?
  - a) Structural hazard
  - b) Control hazard
  - c) Data hazard
  - d) Memory hazard
8. A pipeline encounters delays due to branching instructions, causing incorrect instruction fetching. Which hazard is this?
  - a) Data hazard
  - b) Control hazard
  - c) Structural hazard
  - d) Timing hazard
9. In a system where multiple instructions require the same hardware resource simultaneously, execution is delayed. This is an example of:
  - a) Data hazard
  - b) Structural hazard
  - c) Control hazard
  - d) Logical hazard
10. To reduce pipeline stalls caused by data dependencies, a processor implements forwarding. What does forwarding achieve?
  - a) Reduces instruction size
  - b) Bypasses intermediate storage
  - c) Increases clock speed
  - d) Eliminates memory access
11. A superscalar processor issues multiple instructions per clock cycle. What enables this capability?
  - a) Single execution unit
  - b) Multiple execution units
  - c) Larger memory
  - d) Slower clock
12. In out-of-order execution, instructions are executed in a different order than they appear in the program. What is the main advantage of this technique?
  - a) Reduced memory usage
  - b) Better utilization of CPU resources
  - c) Simplified instruction set
  - d) Reduced instruction count
13. A processor predicts the outcome of a branch instruction and executes instructions ahead of time. This technique is known as:
  - a) Pipelining
  - b) Speculative execution
  - c) Multiprocessing
  - d) Interrupt handling
14. In speculative execution, incorrect predictions may lead to discarded results. What is the main drawback of this approach?
  - a) Increased memory size
  - b) Wasted computation
  - c) Reduced clock speed
  - d) Increased instruction size
15. A processor uses separate units to dispatch instructions to execution units. What is the primary role of the dispatch unit?

- a) Execute instructions
  - b) Fetch instructions
  - c) Assign instructions to execution units
  - d) Store instructions
16. Execution units in a processor handle different types of operations simultaneously. Which of the following is an example of an execution unit?
- a) Cache memory
  - b) ALU
  - c) Control Unit
  - d) Register file
17. A processor with hyper-threading allows multiple threads to share execution resources. What is the primary benefit?
- a) Reduced hardware cost
  - b) Increased instruction throughput
  - c) Lower power consumption
  - d) Smaller memory
18. In simultaneous multithreading (SMT), multiple threads are executed at the same time. What challenge arises from this approach?
- a) Reduced instruction set
  - b) Resource contention
  - c) Lower clock speed
  - d) Reduced memory
19. A system experiences reduced performance when multiple threads compete for the same execution units. This issue is related to:
- a) Instruction pipelining
  - b) Resource sharing
  - c) Cache coherence
  - d) Memory hierarchy
20. A processor pipeline achieves ideal performance when all stages take equal time. What happens if stages are unbalanced?
- a) Improved speed
  - b) Reduced efficiency
  - c) Increased memory
  - d) No effect
21. In a multi-bus system, data can be transferred between different components simultaneously. What is the main advantage of this design?
- a) Simplicity
  - b) Parallelism
  - c) Lower cost
  - d) Reduced memory
22. A processor encounters frequent stalls due to instruction dependencies. Which technique can best improve performance?
- a) Increasing memory size
  - b) Data forwarding
  - c) Reducing clock speed
  - d) Decreasing instruction count
23. A superscalar processor executes instructions out of order but commits them in order. Why is this necessary?
- a) To reduce memory usage
  - b) To maintain program correctness
  - c) To increase clock speed
  - d) To simplify hardware
24. A processor uses speculative execution along with branch prediction. What determines the effectiveness of this approach?
- a) Memory size
  - b) Prediction accuracy

- c) Instruction size
  - d) Number of registers
25. A computing system integrates pipelining, superscalar execution, and SMT. What is the overall goal of combining these techniques?
- a) Reduce hardware complexity
  - b) Increase instruction throughput
  - c) Decrease memory usage
  - d) Simplify programming

***Code of Conduct:***

***I S.Hazipeera (192572056) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.***

***I understand that any violation of academic integrity rules will result in disciplinary action.***

***Signature of the student: S.Hazipeera***

**MARKS ALLOCATION**

|           | <b>Understanding Concepts (25%)</b> | <b>Accuracy of Answers (30%)</b> | <b>Application Knowledge (20%)</b> | <b>Time Management (15%)</b> | <b>Logical Thinking Decision Making (10%)</b> |
|-----------|-------------------------------------|----------------------------------|------------------------------------|------------------------------|-----------------------------------------------|
| <b>Q1</b> |                                     |                                  |                                    |                              |                                               |
| <b>Q2</b> |                                     |                                  |                                    |                              |                                               |
| <b>Q3</b> |                                     |                                  |                                    |                              |                                               |
| <b>Q4</b> |                                     |                                  |                                    |                              |                                               |
| <b>Q5</b> |                                     |                                  |                                    |                              |                                               |

**RUBRICS FOR CALCULATION**

| <b>Criteria</b>                         | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                 | <b>Level 3 – Proficient</b>                                       | <b>Level 2 – Developing</b>                           | <b>Level 1 – Beginner</b>                     |
|-----------------------------------------|------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| <b>Understanding of Concepts</b>        | <b>25%</b>       | Demonstrates clear and complete understanding of concepts tested in MCQs   | Shows good understanding of concepts with minor conceptual errors | Partial understanding; struggles with key concepts    | Misinterprets or fails to understand concepts |
| <b>Accuracy of Answers</b>              | <b>30%</b>       | Selects correct answers for all questions with high accuracy               | Answers most questions correctly with few errors                  | Provides limited correct answers with frequent errors | Unable to select correct answers consistently |
| <b>Application of Knowledge</b>         | <b>20%</b>       | Applies concepts effectively to application-based and analytical questions | Applies concepts with some errors                                 | Limited application of concepts; requires improvement | Unable to apply relevant knowledge            |
| <b>Time Management</b>                  | <b>15%</b>       | Completes all questions within allotted time effectively                   | Completes most questions within time                              | Attempts questions but struggles with time management | Unable to complete test on time               |
| <b>Logical Thinking Decision Making</b> | <b>10%</b>       | Analyzes options carefully and chooses the most appropriate answer         | Makes correct choices based on reasonable analysis                | Shows limited reasoning in selecting answers          | Answers without reasoning                     |

## ANSWERS FOR THE QUESTIONS:

| <b>Q.No</b> | <b>Correct Answer</b>                            | <b>Explanation</b>                                                                                  |
|-------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1           | <b>b) Control Unit</b>                           | The Control Unit controls register transfers and coordinates operations.                            |
| 2           | <b>a) Register Transfer Language (RTL)</b>       | RTL specifies the sequence of data transfers between registers.                                     |
| 3           | <b>b) ALU</b>                                    | The Arithmetic Logic Unit performs logical operations like AND, OR, and XOR.                        |
| 4           | <b>b) Bus contention</b>                         | In a single-bus system, multiple devices cannot use the bus simultaneously.                         |
| 5           | <b>b) Parallel data transfer capability</b>      | Multiple buses allow simultaneous data transfers, improving performance.                            |
| 6           | <b>b) Pipeline stalls</b>                        | A slower pipeline stage delays subsequent instructions.                                             |
| 7           | <b>c) Data hazard</b>                            | Occurs when an instruction depends on the result of a previous instruction.                         |
| 8           | <b>b) Control hazard</b>                         | Branch instructions can cause incorrect instruction fetching.                                       |
| 9           | <b>b) Structural hazard</b>                      | Multiple instructions competing for the same hardware resource.                                     |
| 10          | <b>b) Bypasses intermediate storage</b>          | Forwarding sends results directly to dependent instructions without waiting.                        |
| 11          | <b>b) Multiple execution units</b>               | Superscalar processors execute multiple instructions simultaneously using multiple execution units. |
| 12          | <b>b) Better utilization of CPU resources</b>    | Out-of-order execution keeps execution units busy.                                                  |
| 13          | <b>b) Speculative execution</b>                  | Executes predicted instructions before the branch outcome is known.                                 |
| 14          | <b>b) Wasted computation</b>                     | Incorrect branch predictions require discarding executed instructions.                              |
| 15          | <b>c) Assign instructions to execution units</b> | The dispatch unit sends instructions to appropriate execution units.                                |
| 16          | <b>b) ALU</b>                                    | ALU is an execution unit for arithmetic and logical operations.                                     |
| 17          | <b>b) Increased instruction throughput</b>       | Hyper-threading improves CPU utilization by running multiple threads.                               |
| 18          | <b>b) Resource contention</b>                    | Multiple threads compete for shared CPU resources.                                                  |
| 19          | <b>b) Resource sharing</b>                       | Performance decreases when threads compete for execution resources.                                 |

| <b>Q.No Correct Answer</b> |                                           | <b>Explanation</b>                                                              |
|----------------------------|-------------------------------------------|---------------------------------------------------------------------------------|
| <b>20</b>                  | <b>b) Reduced efficiency</b>              | Unbalanced pipeline stages reduce overall pipeline performance.                 |
| <b>21</b>                  | <b>b) Parallelism</b>                     | Multi-bus systems allow simultaneous data transfers.                            |
| <b>22</b>                  | <b>b) Data forwarding</b>                 | Forwarding reduces stalls caused by data dependencies.                          |
| <b>23</b>                  | <b>b) To maintain program correctness</b> | Instructions commit in order to preserve correct program behaviour.             |
| <b>24</b>                  | <b>b) Prediction accuracy</b>             | Better branch prediction leads to higher speculative execution performance.     |
| <b>25</b>                  | <b>b) Increase instruction throughput</b> | Combining pipelining, superscalar execution, and SMT maximizes CPU performance. |